EXPLORATORY DATA ANALYSIS

CHAPTER 02

Understanding the Dataset

Understand what each row, column, and recorded value actually represents.

A practical, concept-first course for quantitative and visual EDA

Learning objectives

Identify observational units; interpret rows, columns, variables, and values; use data dictionaries and metadata; distinguish analytical variable types from storage types; explain measurement scales and granularity; and document assumptions about the data-generating process.

1. Review of Chapter 1 — What Is EDA?

EDA is an iterative investigation of data before analytical or modeling decisions. Its core questions are: What do I have? Can I trust the data? What patterns or stories does it contain? Quantitative summaries and visualizations complement each other. An observed association is not automatically an explanation or evidence of causation.

2. Review of Chapter 1 assignment

For the customer dataset (Customer ID, Age, Income, Occupation, City, Purchased Product), questions should cover context, structure, quality, distributions, and relationships. The most important structural question is what one row represents. A customer-level table and a transaction-level table support different counts and interpretations.

3. What it means to understand a dataset

A dataset is a representation of something observed, measured, recorded, or generated. A column called Income might mean monthly personal income, annual household income, gross income, or net income. A value becomes interpretable when its definition, unit, time period, and collection context are understood.

4. Observational units

An **observational unit** is the entity about which a record is made: a customer, transaction, student, patient visit, product, household, website session, or day of observations.

Transaction ID	Customer ID	Product	Amount
T001	C101	Laptop	80000
T002	C101	Mouse	1200
T003	C102	Phone	45000

Here each row represents a transaction. Customer C101 appears twice because the same customer made two purchases. Row count, unique-customer count, and total transaction value are different quantities.

Rule: identify the observational unit before interpreting a row count, average, or relationship.

5. Rows, columns, variables, and values

Rows are records; columns are recorded attributes; shape is (rows, columns); a variable is an attribute that can vary; and a value is the recorded result for one variable and observation. Column names are clues, not complete definitions.

```
import pandas as pd\ndf = pd.read_csv("customers.csv")\n\nprint("Shape:", df.shape)\nprint("Rows:", df.shape[0])\npr
```

6. Data dictionaries and metadata

A **data dictionary** documents variable definitions, types, units, permitted values, missing-value conventions, sources, and restrictions. **Metadata** is information about the data: when and how it was collected, where it came from, how values were measured, and what each field means.

Column	Meaning to establish
Customer ID	Identifier; uniqueness depends on row granularity.
Age	Age at what date, and in what units?
Income	Amount, currency, reporting period, personal or household basis.
Occupation	Free text or controlled category list?

Column	Meaning to establish
City	Residence, workplace, transaction location, or another meaning?
Purchased Product	Transaction product, latest purchase, preferred product, or another measure?

When documentation is missing, list ambiguous fields, inspect values and categories, ask the data provider where possible, and record unresolved assumptions. Do not invent definitions.

7. Analytical variable types

Type	Meaning	Examples
Numerical	Quantities for which arithmetic may be meaningful.	Income, weight, elapsed time
Categorical	Groups, classes, or labels.	Occupation, city, product type
Ordinal	Categories with a meaningful order.	Low / medium / high
Date/time	Dates, times, or durations.	Signup date, purchase timestamp
Identifier	Label distinguishing an entity or record.	Customer ID, postal code, product code

Numerical variables may be discrete (counts) or continuous (measurements that can in principle vary across a range). Age in completed years is integer-valued, while age calculated precisely from dates is more naturally treated as continuous. A numeric-looking identifier is not automatically a measurement.

```
df["Customer ID"] = df["Customer ID"].astype("string")
```

8. Measurement scales

Scale	Meaning	Example	Interpretation
Nominal	Categories without inherent order.	City	Compare membership and counts.
Ordinal	Categories with order.	Satisfaction	Order is meaningful; spacing may not be.
Interval	Meaningful differences, no absolute zero.	Temperature °C	Differences meaningful; ratios generally not.
Ratio	Meaningful differences and absolute zero.	Weight, defined income	Ratios can be meaningful.

City codes such as 1 = Dhaka and 2 = Chattogram remain nominal categories; their average has no useful geographic interpretation. Similarly, 20°C is not twice as hot as 10°C.

9. Granularity

Granularity is the level of detail represented by each record. A customer table may have one row per customer; a transaction table may have one row per purchase; a daily sales table may aggregate many purchases into one row per day, store, or product.

Dataset	One row represents	Likely use
Customer table	One customer	Customer attributes
Transaction table	One purchase	Transaction amounts and frequency
Daily sales table	Aggregate for a date and possibly store/product	Trends in aggregated sales

Mixed granularity—such as transaction rows combined with daily totals—can cause double counting or incompatible comparisons. Check identifiers, date precision, row labels, and source definitions.

10. The data-generating process

The data-generating process is how observations were produced and recorded. Ask who or what generated each record, whether data came from a survey, application, sensor, transaction system, or experiment, who could enter the dataset, who might be absent, and whether collection procedures changed.

An online retailer's records may overrepresent customers in a city where advertising was concentrated. The observed distribution describes the recorded data; generalizing to all customers requires evidence about coverage and selection.

11. Practical dataset inspection

```
print(df.shape)\nprint(df.columns.tolist())\nprint(df.dtypes)\ndf.info()\nprint(df.head())\nprint(df.tail())
```

Inspect category frequencies and missingness:

```
print(df["Occupation"].value_counts(dropna=False))\nprint(df["City"].value_counts(dropna=False))\n\nmissing_count = d
```

These checks describe structure; they do not prove that every row is a unique customer or that the data is accurate.

12. Common mistakes

• Assuming one row means one person.

• Treating all numeric-looking columns as measurements.

• Assuming column names fully define variables.

• Ignoring units, time periods, and data definitions.

• Confusing repeated entities with duplicate records.

• Treating recorded data as a complete representation of reality.

• Silently converting assumptions into facts.

13. Chapter summary

Understanding a dataset requires identifying the observational unit, interpreting variables and values, consulting documentation, distinguishing analytical types and measurement scales, understanding granularity, and asking how data was generated. Inspect the structure with code, but document what remains uncertain.

Central principle: understand what the data represents before interpreting what the data says.

14. Assignment

Part A — Conceptual questions

• Define an observational unit and give two examples.

• Why do 1,000 rows not necessarily mean 1,000 unique customers?

• Explain the difference between a data dictionary and metadata.

• Why is Customer ID not usually a numerical measurement?

• Distinguish nominal, ordinal, interval, and ratio scales.

• Explain granularity and the data-generating process.

Part B — Dataset interpretation

Consider a sales dataset with Transaction ID, Customer ID, Product, Quantity, Unit Price, Purchase Date, and City. Identify the likely observational unit; classify the columns; list definitions or units requiring clarification; explain why row count may differ from unique customer count; and state what you would ask the provider before calculating total sales.

Part C — Practical inspection

Using a CSV of your choice, report shape, columns, types, sample rows, unique values, missingness, duplicates, and identifier uniqueness where appropriate. Write at least five observations and three questions that cannot be answered from the values alone.

15. Coding project — Dataset Reconnaissance Notebook

Extend your cumulative EDA master notebook with a reusable inspection section: loading; shape and columns; data types; first and last rows; unique-value counts; numerical and categorical column identification; missingness; duplicate checks; identifier uniqueness where appropriate; and a written overview separating known facts, assumptions, and open questions.

Do not automatically clean, transform, remove outliers, or engineer features. The goal is to understand and document the dataset first.

16. Preview of Next Chapter

Chapter 3 — Data Quality Assessment

Next, investigate completeness, validity, consistency, duplicates, missingness, suspicious values, and the difference between a genuine anomaly and a data-entry error.